# Yggdrasil

**A world-grounded mind for NPCs — capstone project breakdown**

Reference document for COS650 Software Project: Research, Planning, and Design · Area: Artificial Intelligence · Compiled 2026-10-09 from the project vault, hostile-review reports and code on branch rework/2026-09.

How to use this document: it is the raw material for your proposal, business model canvas and white paper. Write the text you submit in your own words, check the AI-use level set on each deliverable, and verify every citation in section 10 yourself before it goes in a reference list. All numbers come from the project's own benchmark and verifier reports as of 2026-09-19 and are reproducible from the repository.

## 1. What Yggdrasil is

Yggdrasil is an artificial-life research engine and desktop laboratory in which simulated people ("beings", internally also "Gahallians") live in a world, learn everything they know from that world, form a culture and pass it on across generations. Each being has a cheap deterministic fast mind that runs every simulation tick, and an LLM slow mind that is consulted only at moments of salience and that reasons solely over what that particular being has experienced or been told. The long-term goal is one brain that can be dropped into any world — a forager continent, a modern town, a fantasy voxel sandbox — and live, learn, adapt and create there the way a human would.

Two framings coexist and both matter for the capstone. As a product it is the NPC brain for the Playrzone platform and the Gahalla game engine: non-player characters that remember, react, decide and grow instead of running scripts. As research it is a controlled, replayable instrument for studying how knowledge, technology, institutions and belief emerge in a population of minds.

## 2. Research question

> *Can an NPC mind whose knowledge is grounded entirely in its world — no scripted behaviour, no pretrained facts — survive, adapt and transmit culture better, and more believably, than a plain frontier-LLM agent; and does the same mind work unchanged across different worlds?*

Sub-questions:

- Does world-grounded learning (learned affordances + place memory + curiosity) beat an authored survival script on objective survival and adaptation metrics?
- How much pretrained knowledge leaks from the LLM into beings, and can it be measured reliably?
- Is the mind truly world-agnostic — can a second world be added with a zero-line change to the mind?
- Do blind judges prefer traces from the grounded mind over a plain LLM agent, and does that preference survive pre-registered statistics?

## 3. Doctrine — the design principles that make it different

1. **The world is the only source of knowledge.** Deterministic, interpretable "miners" are the only writers of semantic knowledge, place memory, skills, person models and the life story. False beliefs are legal and protected.
2. **The LLM reasons; it does not know.** The slow mind receives only what the being holds and may produce goals, plans, speech, names and hypotheses to test — never facts. A hypothesis becomes knowledge only when the world confirms it through a miner.
3. **Nothing is scripted.** No rule says "explore", "hunt" or "teach your child". The mind gets general machinery; the world supplies reasons.
4. **World-agnostic by construction.** Mind code is written against drives, kind tags, event roles, a dozen core verbs (Move, Build, Take, Use, Combine, Wait, …) and an opaque notion of place — never world nouns and never coordinates. This is enforced by compiler-level proofs, not by convention.
5. **Deterministic and replayable.** Same seed → bit-identical run. Every slow-mind decision is recorded losslessly with its prompt, raw reply, parse, the version the world admitted, and the outcome it led to.
6. **Honest reporting.** Negative results are results; the analysis layer never feeds back into the simulation; nothing is tuned to make a number pass.

## 4. Architecture

| Layer | Role |
|---|---|
| L0 WorldAdapter | What a world is: kinds and their tags, actions bound to core verbs, drives, context features, a SpaceOps interface (distance / direction / within / equal over opaque PlaceRef handles), and a lexicon. Implemented today by the forager world and a toy graph world; a modern town and the Gahalla voxel world are next. |
| L1 Body | Needs and drives, perception, action execution. A world gate validates every intent against the world's declared actions; refusals are returned as values the being remembers. |
| L2 Fast mind | Runs every tick for thousands of beings at zero inference cost. Today an authored reflex policy that lives on the world side; next, a learned arbiter (Stage 4) that scores options from the being's own experience. |
| L3 Slow mind | LLM (gpt-oss-120b via Fireworks today; a v0 fine-tune exists), salience-triggered, JSON-schema output, behind a pluggable IMindBackend interface with mock and scripted backends for testing. |
| L4 Memory | Episodic stream with a bounded footprint, consolidation, and self-describing provenance levels (off / compact / full). |
| L5 Miners | OutcomeMiner, SocialMiner, CraftObservationMiner (watch-to-learn), SuccessionMiner, ContextMiner — deterministic learners that turn episodes into rules with confidence, evidence and contradiction counts. |
| L6 / L7 Identity & society | Kinship, relationships, narrative, institutions; the population-level structures that emerge from individual minds. |

| Layer | Role |
|---|---|
| Instrument | Cognition recorder → corpus (JSONL), dataset card, run database, Electron lab UI, and an MCP (Model Context Protocol) server exposing the lab as tools. |

### One simulation tick, end to end

7. The body updates drives (hunger, rest, safety, …) and the world advances (food, weather, animals, other beings).
8. Perception builds a ThinkInput: what is around, how the being feels, and what it holds — memories, knowledge, relationships — rendered through the world's lexicon.
9. The fast mind emits a plan (a list of intents over core verbs) at no cost.
10. If something salient happened or a drive is pressing, the slow mind is prompted with what the being holds and returns goals, a plan, utterances and hypotheses as JSON.
11. The gate validates every intent; the body executes; events fire for witnesses.
12. Miners fold the new episodes into knowledge, place memory and relationships.
13. The recorder writes the full decision record linked to its outcome; the lab and dataset card read from it; nothing reads back into the sim.

## 5. Tools and stack

| Area | What we use |
|---|---|
| Language / runtime | Strict TypeScript, Node, pnpm workspace monorepo (packages: engine, lab, home, db, mcp). ESM with no DOM side effects at engine load, so the engine runs headless in Node, in a Web Worker, and in Electron. |
| Lab (the instrument) | Electron + React + three.js (WebGL). Simulation runs in a Web Worker; semantic-zoom bands (FAR / DEFAULT / MID / NEAR), true scale (1 unit = 1 m), colour-blind-safe condition encoding, roster with filter-highlight, hover cards, annotation chips, window.__ygg ground-truth probes, a health probe (shader / console / window errors), 60 / 10 / 2 fps frame limiter. |
| LLM | Fireworks API with gpt-oss-120b as the slow mind; fine-tuning on Fireworks (v0 done); mock and scripted backends for every unpaid run; a hard-capped CallBudget enforced in code (store-authoritative, refuses call N+1, human-only override file). |
| Testing & proof | Vitest with fake clocks; golden state fingerprints (SHA-256); byte-identity dumps of old vs new engine in clean checkouts; differential parser fuzzing (~29,000 replies); TypeScript-compiler-API guards (a mind must compile with the worlds/ directory deleted; a type-graph walk proves nothing a mind is constructed with can reach coordinates); runtime leak-route tests; Playwright/CDP driving the real Electron app with real input events. |
| Data | Cognition records (schema v3) as JSONL under ~/.yggdrasil-lab; dataset card generator; replay tooling; blind evaluation sets kept outside the repository. |

| Area | What we use |
|---|---|
| Tooling | MCP server with tools such as make_conscious, dataset_card, knowledge_fidelity, run_ab, judge_traces, ignorance_probe; Obsidian vault for decisions, architecture and the progress log; git with local branches; Python/matplotlib for earlier figures. |
| Destination engine | Gahalla — a TypeScript voxel game engine — through a headless GahallaWorldService (terrain, line of sight, km-scale path-finding, building through the same edit seam human players use). |

# 6. How we work (methodology)

## Research method

- Every behaviour change is a hypothesis with an ablation switch, new goldens recorded once, an objective world metric measured in a seed-paired A/B, and a mock-backed blind trace comparison; paid LLM runs follow a written budget decision.
- Objective metrics first (survival, diet breadth, discovery and its spread, project coherence), blind judgments second; both reported with uncertainty.
- Negative results are written up with the same care as positive ones — several of the most useful findings so far are negative.

## Engineering process

- An architect plans, decomposes work into items with explicit file fences, reviews every diff and directs revisions; implementation is carried out by AI coding agents (Claude Opus) under tight briefs.
- Every item is then attacked by a separate hostile verifier that uses its own seeds, configurations and adversarial inputs, re-derives reported numbers from raw events, plants bugs to prove the tests bite, and ranks what it finds. Anything visible also gets a visual critic that scores screenshots blind against ground truth. "Acceptable" is a fail; items loop until the verifier passes.
- Decisions are written to the vault before implementation; every delivered item is checkpoint-committed; work stays on a local branch until reviewed.

## Behaviour-neutral refactoring as a discipline

- Structural changes to the mind (moving tables, ids, actions and geometry out of it) must keep state fingerprints, ordered events, prompts and corpus rows byte-identical to the previous engine. Verifiers dump both engines on unseen configurations and diff them; the record schema version is pinned.
- World-agnosticism is proven, not asserted: compile-time proof that the mind builds without any world, a structural import-closure proof that no codec or coordinate type is reachable from mind code, and a runtime test that thirty reflection / cloning / inspection routes yield nothing from a place handle.

# 7. What has been built and found so far

## Research findings (experiment arc, June 2026)

- **Civilisational substrate emerges unscripted.** Over fourteen experimental cycles, settlements, crafts and institutions appeared without any rule asking for them; impersonal institutions outlived their founders while personal authority did not.
- **Conscious minds inherit culture immediately.** Beings with LLM slow minds adopted existing practice at once; followership was relational rather than meritocratic; stated "meaning" was instrumental.
- **The bottleneck on discovery was culture and conditions, not the mind** (curiosity arc).
- **Myths create destinies; story moves faster than practice** (experiments 10–11 and the history arc).
- **Technological inheritance, experiment 1 (apprenticeship): strong positive.** A watch-to-learn miner took a craft from 1 maker to 168 makers over six generations.
- A corpus of ~2,120 recorded acts was curated and a v0 fine-tune trained; the lesson recorded at the time was that training is not the milestone — understanding is.

## Engineering results (rework campaign, September 2026)

- **Correctness and scale.** A uniform-grid spatial index took the world from 9.3 to 408 ticks/s at 1,000 beings (independent benchmark; suite 153 s → 9 s). Memory was bounded: 2.63 million episodic records (~1.76 GB) at day 30 with 256 beings became ~97 MB, bounded by the living population.
- **Corpus v2/v3.** Lossless, replayable, outcome-linked cognition records with a scaffold ledger for interventions: 0 replay mismatches over 5,500 rows; 1,765 outcomes rebuilt independently with 0 mismatches.
- **Knowledge-fidelity measurement.** A tiered leak detector (structural, lexical, LLM judge) with a blind calibration card: the lexical tier is high-precision (0.98) and low-recall (0.51–0.68) by design — a lower bound; paraphrase and translation leaks require the judge tier.
- **True-scale world.** An 8,192 m forager continent (peak 899 m, carrying capacity, respawn out of sight) added as adapter data with an empty diff over the mind. Key finding: beings starve within 500 m of the birth valley while ~14,000 food items sit 2–4 km away; reflex minds never explore or hunt. This is the evidence that drives the Stage 4 mind.
- **The WorldAdapter seam (B1–B4).** Tables, ids, actions and geometry were moved out of the mind, each step byte-identical. Places are now opaque at runtime (a per-world class with private fields; a second world's places are refused), the decoder is a capability handed only to the engine's edges, and the verifier's own seven-configuration harness shows 0 of 7 differences. One blocker remains: ~7% slower at 1,000 beings against a 3% budget.
- **Lab.** Simulation in a Worker (0 long tasks over 5,757 frames at maximum speed); beings visible and countable (blind counts within 3%; 474 hover offers with 0 mispicks while paused; roster 42/42 real-mouse clicks; unwell beings findable cold); security hardening of the IPC and key handling.
- **Gahalla world service (M2).** Headless terrain / line-of-sight / path-finding / building service: density sample 40 ns, $32^3$ chunk mesh 2.2 ms, cold navigation page 0.4–0.7 ms, 300-agent tick 1.3 ms median. The verifier found a 111 ms worst case caused by one unbounded path search; a budgeted, resumable search is in progress.
- Test suites at the last buildable commit: engine 927, lab 499, MCP 37, all type-checks clean.

## 8. Evaluation and validation design

This is the part of the project that was reviewed hardest (five hostile methods reviews) and the part a capstone committee will care most about.

- **Arms.** A = the full Yggdrasil mind; A− = ablations (no epistemic instruction, no curiosity, no place memory, …); B = a plain frontier-LLM agent with its own memory, built as a fair generative-agents-style baseline; B+ = the baseline with Yggdrasil's epistemic instruction.
- **Design.** Seed-paired runs; the unit of analysis is the seed (a cluster), not the pair — pairs from one run are correlated, which an early version of the harness got wrong and the reviewer caught (type I error up to 0.93 at a correlation of 0.5; the cluster-level gate brings it to 0.003).
- **Blinding.** Traces are relabelled with handles drawn from a per-run secret; the manifest carries only a commitment; unsealing is logged once and any judging after unsealing is stamped non-blind and unusable.
- **Pre-registration.** The judge gate ($\alpha = 0.05$, declared effect, assumed tie rate, minimum decisive clusters) is frozen in code; the verdict path accepts no override; an exploratory path exists but can never print a usable verdict. Every artefact recomputes power at the observed tie rate and reports itself underpowered when it is.
- **Metrics.** Objective: starvation deaths with food within 2 km, diet breadth over a life, first discovery of hunting and its spread by observation, plan success rate, population over time. Epistemic: knowledge-fidelity leak rate (three tiers), ignorance-probe separation (does a being disavow what it cannot know, without hedged leaks). Subjective: pairwise blind judge preference ("which trace reads as more alive"). Integrity: refusal, idle and silent-fallback rates per arm; a degraded arm makes the run unusable.
- **Inference.** Exact sign-flip permutation test and sign test over seeds, both required; Holm correction across the declared family of contrasts; a minimum number of seeds below which no verdict prints.
- **Spend control.** A hard cap of 2,000 model calls enforced in code. The first paid run is estimation-only (judge tie rate, both-orders consistency, between-run correlation, calls per being, detector calibration) at ~1,130 calls and can never print a verdict; a confirmatory design is then computed from the measured values (currently ~2,800 calls) and needs explicit approval.
- **Threats to validity that are now measured rather than assumed:** pseudo-replication, judge position bias and ties, silent LLM fallback to reflex (now counted at the source), lexical detector recall, mock arms that do not actually act (integrity gate), author bias in adversarial test sets (verifiers write their own).

## 9. Data — the cognition corpus

- **Original data, self-collected.** Every slow-mind decision becomes a cognition record: the ThinkInput (what the being held), the composed prompt, the raw model reply, the parsed output, the output the world admitted, the outcome it led to, any intervention (scaffold ledger) and the provenance level. Schema version 3.
- **Collection methodology** is documented in code and tests: rows are lossless (a stored row re-serialises to the same bytes), replayable (the prompt can be rebuilt from the row alone with the world), and outcome-linked (0 mismatches when outcomes are rebuilt independently).

- **Dataset card** reports counts, the validity definition (an empty decision is not a decision; prose salvage cases are explicit), leak rate with its blind calibration, outcome-linked rate, recorded interventions, and the provenance level used.
- **Curation** handles duplicates and near-duplicates and balances the corpus; blind evaluation sets are kept outside the repository and never shown to engineers.
- **Purposes:** evaluation (section 8) and, later, distillation of a learned fast mind from slow-mind decisions (decision D3: only after the Stage 4 faculties beat the baseline).
- Exploratory analysis you can do today on any run: leak rate by domain, think cadence, plan success by action kind, outcome-linked rate, and how knowledge, place memory and relationships grow over a life.

## 10. Related work and comparison products

Anchors for the literature review (verify each citation before use):

- Park et al., 2023, "Generative Agents: Interactive Simulacra of Human Behavior" — LLM agents with a memory stream, reflection and planning in a small town. The closest ancestor. Difference: their agents know the world through pretrained knowledge; Yggdrasil's learn it and are measured for leakage.
- Wang et al., 2023, "Voyager: An Open-Ended Embodied Agent with Large Language Models" — lifelong learning in Minecraft with a growing skill library; relevant to the planned skills faculty (F5).
- Altera, "Project Sid" (2024) — roughly a thousand LLM agents forming societies in Minecraft; relevant to scale and emergent institutions.
- Epstein & Axtell, 1996, "Growing Artificial Societies" (Sugarscape); Ray's Tierra; Avida — agent-based and artificial-life models of emergent social structure, the research tradition Yggdrasil's experiment arc belongs to.
- Cognitive architectures: ACT-R (Anderson) and Soar (Laird) — declarative vs procedural memory, chunking; the fast/slow split and the miners echo these.
- Dual-process theory (Kahneman, "Thinking, Fast and Slow") as the framing for a zero-cost fast mind and a salience-triggered slow mind.
- Cultural evolution: Boyd & Richerson; Mesoudi — transmission chains and fidelity, the basis of the knowledge-fidelity battery and the apprenticeship experiment.
- Classic game AI — behaviour trees, GOAP, utility AI (as shipped in Unreal and Unity) — the engineering baseline: scripted, cheap, cannot learn.
- Commercial LLM-NPC products — Inworld AI, Convai, NVIDIA ACE — dialogue-first, pretrained knowledge, inference cost on every line; the market comparison.
- World-model agents (e.g. Hafner et al., Dreamer) as adjacent work on learning from a world rather than from text.

## 11. Business model canvas — sketch

| Block | Content |
|---|---|
| Unique value | NPCs that genuinely learn from their world instead of running scripts or reciting pretrained facts; one portable brain that works across worlds through a declared WorldAdapter contract; deterministic and replayable (a research instrument as well as a product); cheap at scale — the fast mind is free and the LLM is consulted only at salient moments. |
| Comparison | Inworld / Convai / NVIDIA ACE: dialogue-centred, pretrained knowledge, per-line inference cost. Behaviour trees / GOAP / utility AI: cheap but cannot learn or surprise. Generative-agents research prototypes: expensive, not grounded, not deterministic. |
| Customers | Playrzone players and creators; game studios wanting living NPCs (SDK); A-life and social-simulation researchers (instrument, datasets). |
| Monetization | Playrzone platform revenue (creator economy, PlayrCoins); NPC-brain SDK and licensing for other engines via the adapter contract; hosted being-hours; research datasets and licensing. |
| Key metric | Cost per being-hour at a given believability score; leak rate; retention of players around living NPCs. |
| Costs | Slow-mind inference, engineering, evaluation runs; mitigated by salience triggering and a distilled fast mind. |
| Channels, relationships, partners | To be filled in by you: Playrzone community, engine marketplaces, research collaborations. |

## 12. Current state and limitations (honest)

- Branch rework/2026-09 holds more than forty verified commits plus several unfinished work-in-progress items: the B4 performance fix, lab round 7 (an automated acceptance harness and a redesign of how occupied shelters are drawn), Gahalla M2 round 2, and evaluation-harness round 6.
- No paid evaluation run has happened yet; every gate is proven on mock backends only. The estimation pilot is designed and costed but waiting on approval and an API key.
- Only one real world exists (plus a toy graph world used for proofs); the modern town — the real proof of "one brain, any world" — is not built yet.
- The fast mind is still an authored script; beings cannot yet hear each other (F6); goals do not persist across thinks (F3); nothing is forgotten (F4); there is no affect, self-model or theory of mind yet (F7–F9).
- The lexical leak detector's recall is 0.5–0.68; the LLM judge tier is unproven on real data.
- Performance: ~7% regression at 1,000 beings after the geometry seam; the lab holds 60 fps / 0 long tasks in most bands but has regressed twice under review.
- World realism: shelters have no capacity (63 beings were observed in one 4.4 m hut); shelter capacity belongs in adapter data and is noted for the next behaviour-change item.

- Bias and threats: the slow-mind model's pretrained knowledge; judge-model bias; single-machine timing noise (Windows CPU time is quantised at ~15.6 ms, so short benchmarks are noise); author bias in adversarial sets, mitigated by independent verifiers.

## 13. Roadmap and future goals

### Stage 2 — finish the seam

- B4 performance within budget → B5 lexicon (places and distances get words, closing the last textual leak route) → B6 instrument per world → W1-C modern town world with an empty diff over the mind.

### Stage 3 — the world

- M1 viewer: terrain LOD and streaming for the 8 km continent. M2: verify and merge the Gahalla world service. M3: beings living in the Gahalla voxel sandbox — building real voxel huts with the same edit deltas players use, a human avatar walking beside them. M4: crowd rendering for 1,000 beings.
- Final acceptance test: a 1.7 m being walks through the 2 m doorway of a hut it built, beside a 12 m tree and an 800 m ridge, with a human avatar next to it, 300 beings at ≥ 55 fps.

### Stage 4 — the mind (ten faculties, in order)

- F1 fast mind v2: drive arbitration over learned affordances with an uncertainty bonus (curiosity as a property of the arbiter, not a script) — replaces the authored policy. F2 place memory: a miner-built cognitive map with told-but-unverified places and an exploration frontier.
- F3 goals and projects that outlive a think. F4 forgetting, consolidation and sleep (bi-temporal knowledge: "what I used to think"). F5 skills (procedural memory mined from the being's own successful plan fragments, taught by demonstration).
- F6 hearing — speech reaches other minds as testimony that can create hypotheses, never knowledge; contamination measured, not censored. F7 affect. F8 self-model and life story. F9 theory of mind. F10 development (childhood, senescence).
- What "alive" will mean when Stage 4 is done: put a being in a world it has never seen; with no script and no pretrained facts it gets hungry, tries things, learns what feeds it and where, remembers and returns, wanders further when home runs out, watches others and picks up what they can do, asks someone who might know and hears the answer, commits to building something and finishes it next week, grieves a death and avoids the place for a while, can tell you who it is and what happened to it — and is still recognisably the same person when it is old. Then the same mind, unmodified, does all of that in a town.

### Stage 6 — evaluation and training

- Estimation pilot → confirmatory blind A/B → dataset card and white paper → distilled fast mind (decision D3) → human participation (players and researchers inside the world).

## 14. Recommended capstone scope and rubric mapping

Recommendation: make the capstone **"A world-grounded learning mind for NPCs: learned affordances and place memory (Stage 4, F1 + F2), evaluated against an authored reflex policy and a plain-LLM agent in a blind, pre-registered, seed-paired A/B."** It is independently evaluable, has two baselines, uses the evaluation harness that already exists, fixes the single most damning measured finding (starvation beside plenty), is measurable the day it lands, and maps cleanly onto the AI/ML rubric. Everything else in this document is context and future work.

| Rubric requirement | What exists today | What the capstone adds |
|---|---|---|
| Business model canvas | Product direction (Playrzone / Gahalla), comparison list, cost model via the call budget. | Write the canvas: unique value, comparisons, monetization (section 11). |
| Literature review | Doctrine and audit reports cite the design space. | Formal review across generative agents, embodied LLM agents, A-life, cognitive architectures, cultural evolution and game AI (section 10). |
| Data sources and collection | Original data: cognition corpus v3 with dataset card; collection methodology documented in code and tests. | Write up the methodology; run exploratory analysis on a baseline corpus (leak rate, think cadence, outcome-linked rate, knowledge growth). |
| AI/ML MVP | Engine + lab + MCP; multiple backends (gpt-oss-120b, v0 fine-tune, mocks); authored fast mind. | The learned fast mind (F1 + F2) as a new, clearly identifiable component behind an ablation switch. |
| Model evaluation and validation | Blind seed-paired A/B harness, leak battery, ignorance probe, pre-registered gate, spend cap — all proven on mocks. | Run the estimation pilot, then the confirmatory A/B; report metrics, power, limitations and bias honestly. |
| Compare models / approaches | Arms A / A− / B / B+ designed. | Authored reflex vs learned fast mind vs plain-LLM agent, plus ablations (curiosity off, place memory off, observation learning off). |
| Optimization / improvement | Spatial index, memory bound, performance budgets, fine-tune v0. | Document them; if time allows, distil a fast mind from the corpus and compare. |
| Visualizations | Lab world view, dataset card, earlier matplotlib figures. | Custom charts: survival curves, diet breadth, discovery and spread per generation, leak rates with intervals; a time axis in the lab. |
| Deployment / demonstration | Electron lab, MCP server, release pipeline (auto-update). | Packaged lab build plus a recorded end-to-end demo and a headless command that runs the A/B. |

| Rubric requirement | What exists today | What the capstone adds |
|---|---|---|
| White paper / thesis | Vault: architecture, directives, design briefs, five audits, twenty-plus review reports, progress log. | Write it from these, including the negative results. |

### A twelve-week shape

1. Weeks 1–2: proposal, canvas, literature review; finish B4 performance; baseline corpus and exploratory analysis.
2. Weeks 3–6: F1 + F2 behind the ablation switch, byte-neutral when off; new goldens; objective metrics on the continent and the toy world.
3. Weeks 7–8: hostile verification; estimation pilot (capped); confirmatory design from measured numbers.
4. Weeks 9–10: confirmatory A/B (if approved) or mock-backed full battery; charts; dataset card.
5. Weeks 11–12: white paper, packaged demo, limitations and future work.

## 15. Notes on AI use and authorship

- The capstone requires your contribution to be substantial, clearly identifiable and independently evaluable. In this project you own the direction, the doctrine, the research questions, the decisions and the acceptance criteria; AI coding agents were used as an implementation workforce under review, and independent AI verifiers attacked every item. Describe that process accurately in the white paper and follow the course's disclosure rules.
- Each deliverable carries its own AI-use level; the "Choose an Area" activity was Level 1 (no AI). Check the level before using any generated text, and write submitted prose yourself.
- The vault, the review reports and the git history are your evidence of process: what was decided, what failed, what was measured and by whom.